

Curriculum vitae

Address

Yasra Chandio
307 Knowles Engineering Building, UMass Amherst
E-mail: ychandio@umass.edu, Homepage: <https://yasrachandio.github.io/>

Areas of Specialization

I explore the fundamental problems in **Mixed Reality** platforms and applications that impact the physical or cognitive safety of a user.

Education

01/2020–present	PhD in Computer Engineering, University of Massachusetts Amherst
09/2018–12/2019	MS Computer Science, Binghamton University (SUNY)
01/2010–02/2014	BE Software Engineering, MUET, Pakistan

Employment

01/2020–present	Graduate research assistant, Emerging Technologies Lab , UMass Amherst, advised by Fatima M. Anwar.
09/2018–12/2019	Graduate research assistant, Computer Science Department, Binghamton University, advised by Aditya Mishra and Gaunhua Yan.
01/2016–06/2017	Research associate, Systems and Networks Lab , LUMS, Pakistan, advised by Muhammad Hamad Alizai.
12/2014–12/2015	Research engineer, Systems and Networks Lab , NUCES, Pakistan, advised by Affan A. Syed.

Research Projects

Mixed Reality Security, (UMass Amherst)

Mentor: Fatima M. Anwar

I am exploring a new attack surface in Mixed Reality (MR) that is resilient to state-of-the-art sensor fusion techniques. I Leverage machine learning and data-driven heuristics to devise and evaluate stealthy attacks, by concurrently manipulating multiple sensing streams across multiple dimensions.

Quantifying Presence in Mixed Reality (UMass Amherst)

Mentor: Fatima M. Anwar

I am working on devising an objective, unbiased, and real-time system-level metric to quantify the level of presence for a user in mixed reality to enhance the cognitive safety of the users.

Employing Distributed Energy Storage for Grid Peak Reduction (Binghamton University, NY)

Mentor: Aditya Mishra

Leveraged distributed energy storage to enhance the efficiency and reliability of electric grid. Using distributed storage, our algorithms reduced grid-wide peak by 23% and generation costs by 14%.

Retrofitting HVAC for Personalized Comfort in Developing Countries (LUMS, Lahore)

Mentor: Muhammad Hamad Alizai

Designed Hawadaar, a system that retrofits legacy air conditioning devices to achieve HVAC like control in developing countries. Worked on data acquisition, processing, and analytics from devices.

Energy Stack: Managing Energy in Embedded Systems (NUCES, Islamabad)

Mentor: Affan A. Syed, Muhammad Hamad Alizai

Developed a stack for energy management and wireless energy transfer. Designed and evaluated energy scheduling algorithms and code instrumentation methods for checkpointing.

Academic Honors

2022	Best Ph.D. Forum Presentation Award at ACM SenSys.
2022–present	Computing Research Association-Education (CRA-E) Graduate Fellow at CRA.
2022	Three Minute Thesis (3MT) finalist at UMASS Amherst. [Talk]
2022	Scholar at CRA-WP Grad Cohort.
2021	Invited to attend Google PhD Fellowship Summit.
2021	Honorable mention in UMASS ECE 3MT.
2021	Selected for Computer Research Mentorship (CSRMP) Program at Google.
2021	Scholar at CRA-WP Grad Cohort.
2019	Grace Hopper Scholar.
2019	NSF travel grant to attend IEEE IGSC in Alexandria, VA, USA.
2019	Summer Provost Fellowship at Binghamton University.
2017	Best Paper Audience Choice Award at ACM BuildSys.
2017	ACM SenSys travel grant to attend ACM BuildSys/SenSys in Delft, NL.

Teaching and Mentoring

Fall 2021-22	Mentoring graduate students for course projects, Networked Embedded Systems Design (ECE 597SD). Projects on: (a) Deep learning based Sensor Fusion for Mixed Reality. (b) Teleportation into Mixed Reality World. (c) Haptics Illusion in Mixed Reality. (d) Pose Estimation for Micro Movements in Mixed Reality. (e) Remote collaboration using Virtual Reality.
Summer 2022	Instructor, UMASS Turing Summer Program. Conducting lectures on the topic of <i>Elements of Programming</i> and Demo on <i>Extended Reality Applications in Health Care and Beyond</i> .
Spring 2022	Teaching Assistant, Introduction to Embedded System (ECE 231).
Summer 2021	Instructor and coordinator, 3-weeks UMASSS K-12 Engineering Outreach Workshop. Hands-on labs on <i>Getting Started with Embedded Systems</i> .
Spring 2021	Teaching Assistant, Introduction to Embedded System (ECE 231).
Spring 2020	Guest lecturer, Introduction to Embedded System (ECE 231) on topic of <i>BeagleBone Black</i> .

Publications

- [1] **Yasra Chandio**, Noman Bashir, Fatima M. Anwar, *Dataset: HoloSet - A Dataset for Visual Inertial Pose Tracking in Extended Reality*, 5th International SenSys/BuildSys Workshop on Data (**DATA**), 2022. **[PDF]**
- [2] **Yasra Chandio**, *Towards Secure and Immersive Mixed Reality: PhD Forum Abstract*, 20th ACM Conference on Embedded Networked Sensor Systems (**ACM SenSys**), 2022. (**Best Presentation Award**) **[PDF]**

- [3] Khotso Selialia, **Yasra Chandio**, Fatima M. Anwar, *Federated Learning Biases in Heterogeneous Edge-Devices - A Case-study*, at 4th International SenSys/BuildSys Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (**AIChallengeIoT**), 2022. [[PDF](#)]
- [4] **Yasra Chandio**, Noman Bashir, Victoria Interrante, Fatima M. Anwar, *Investigating the Correlation Between Presence and Reaction Time in Mixed Reality*, **under review** at IEEE Transactions on Visualization and Computer Graphics (**TVCG**). [[PDF](#)]
- [5] **Yasra Chandio**, Noman Bashir, Fatima Anwar. *Exploring Stealthy Spatiotemporal Attacks in Extended Reality*, **under review** at ACM/IEEE International Conference on Internet of Things Design and Implementation (**IoTDI**), 2023. [[PDF](#)]
- [6] Noman Bashir, **Yasra Chandio**, David Irwin, Fatima Anwar, Jeremy Gummeson, Prashant Shenoy. *Acclimator: Jointly Managing Electrical and Thermal Energy in Environmentally-powered Systems*, **under review** at (**ACM SIGMETRICS**), 2023. [[PDF](#)]
- [7] **Yasra Chandio**, Fatima M. Anwar, "Poster: Spatiotemporal Security in Mixed Reality Systems", In The 18th ACM Conference on Embedded Networked Sensor Systems, (**ACM SENSYS**), 2020. [[PDF](#)]
- [8] **Yasra Chandio**, Aditya Mishra, Anand Seetharam, "GridPeaks: Employing Distributed Energy Storage for Grid Peak Reduction", IEEE International Green and Sustainable Computing conference, (**IEEE IGSC**), 2019. [[PDF](#)]
- [9] **Yasra Chandio**, Jo A gila Bitsch, Affan A. Syed, Muhammad Hamad Alizai: "Networking Wireless Energy in Embedded Networks", ACM Transaction on Sensor Networks (**ACM TOSN**), 2018. [[PDF](#)]
- [10] Samar Abbas, Abu Bakar, **Yasra Chandio**, Khadija Hafeez, Ayesha Ali, Tariq M Jadoon, Muhammad Hamad Alizai, "Inverted HVAC: Greenifying Older Buildings, One Room at a Time", (**ACM TOSN**), 2018. [[PDF](#)]
- [11] Khadija Hafeez, **Yasra Chandio**, Abu Bakar, Ayesha Ali, Affan A. Syed, Tariq M. Jadoon, Muhammad Hamad Alizai: "Inverting HVAC for Energy Efficient Thermal Comfort in Populous Emerging Countries", ACM International Conference on Systems for Energy-Efficient Built Environments (**ACM BuildSys**), 2017. (**Audience choice award winner.**) [[PDF](#)]
- [12] M. Hamad Alizai, Qasim Raza, **Yasra Chandio**, Affan A. Syed, Tariq M. Jadoon: "Simulating Intermittently Powered Embedded Networks", 13th International Conference on Embedded Wireless Systems and Networks (**EWSN**), Graz, Austria, 2016. [[PDF](#)]

Service to the Profession

Shadow PC member for ACM SenSys 2022.

Shadow PC member for ACM EuroSys 2022.

Review committee member for Extended Reality track at Grace Hopper Celebration 2022.

Review committee member for VR/AR/MR track at Grace Hopper Celebration 2021.